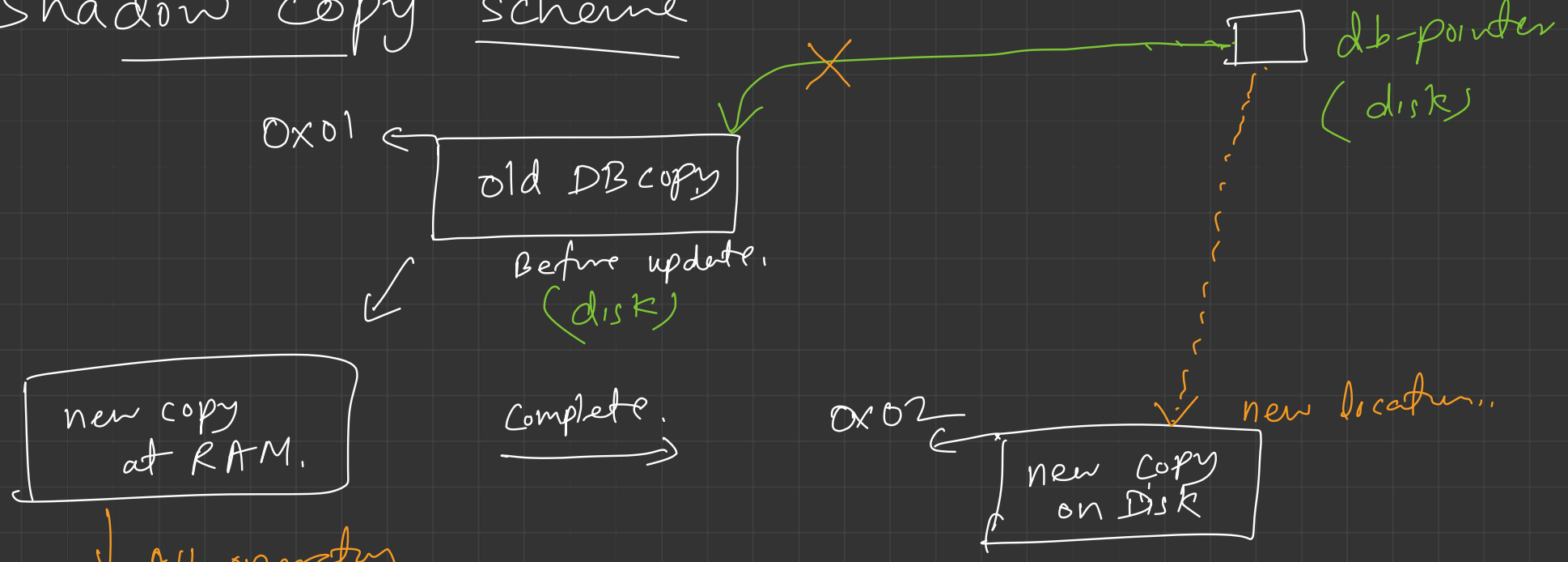
Lec-13

* How to implement Atomiticity & Durability?

① Shadow copy scheme



db-pointer updated to
new disk location
→ notify Txn commit

Log Based Recovery :-

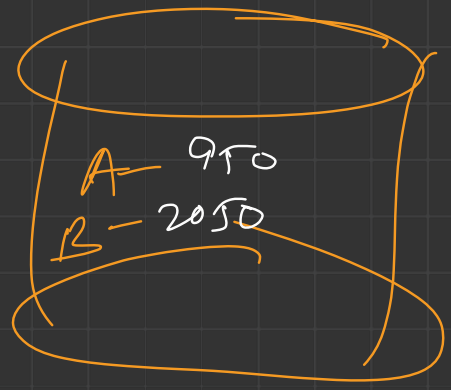
1

A	B
₹1000	₹2000

T_0:
read(A)
A := A − 50
write(A)
B := B + 50
write(B)

Logs (stable storage)

<T_0 start>
<T_0, A, 950>
<T_0, B, 2050>
<T_0 commit>



① Deferred DB Modification

② Immediate DB modification

<T_0 START>

<T_0, A, 1000, 950>

<T_0, B, 2000, 2050>

<T_0 Commit>

old item value

new item value.

→ Checkpoints

$C_1 \rightarrow$ $\Sigma T_n \rightarrow$ commit.

$C_2 \rightarrow$ ΣT_n cont.

H.W → checkpoints

Google

Commit